+1 (437) 989-8026
Greater Toronto Area
catherineyq.chen@mail.utoronto.ca
LinkedIn Page | Personal Page

# Catherine Chen

## Summary of Skills and QualiFications

- Master's candidate in Financial Insurance with credit risk (EAD/CCF), longevity risk, and Monte Carlo modeling experience
- FRM Part I candidate with skills in financial analysis, quantitative risk, statistics, and machine learning
- Proficient in Python, R, SQL, MATLAB, Google Analytics, Excel/VBA, and Power BI for data analysis and automation

## Education

- **Candidate, Master of Financial Insurance (MFI)** Sep. 2026 – Present (Expected completion 2027)
  University of Toronto, Department of Statistical Sciences
  Relevant Courses: Mathematical Finance, Insurance Risk Management, Data Science and Machine Learning

- **Bachelor of Mathematics, Co-op Program (With Distinction)** Sep. 2021 – Jun. 2026
  University of Waterloo, Department of Statistics and Actuarial Science
  Honours Mathematics/Financial Analysis and Risk Management (FARM)

## Research & Technical Projects

- **American Put Option Pricing and Early Exercise Analysis, University of Toronto**
  *Toronto, Canada* Nov. 2026
  • Implemented **CRR binomial pricing in Python (NumPy)** and derived variational **PDE conditions** for American put options
  • Analyzed **early exercise boundaries and delta hedging**, testing price convergence across interest-rate and volatility scenarios
  • Evaluated discounted P&L, exercise timing, and volatility misspecification using **Monte Carlo simulation** with 10,000 paths

- **Exposure at Default and Credit Conversion Factor Analysis, University of Toronto**
  *Toronto, Canada* Oct. 2026
  • Cleaned and integrated six source tables in **Python (pandas)**, building an analytical dataset of 4,733 revolving default episodes
  • Calculated **EAD and CCF** from historical balances, credit limits and pending authorizations using a 180-day observation horizon
  • Implemented **data quality controls and reconciliation checks** to validate record timing, episode IDs and EAD/CCF calculations

- **Mortality Pooling and Longevity Risk in Life Annuity Pricing, University of Toronto**
  *Toronto, Canada* Jul. 2026 – Sep. 2026
  • Compared individual and pooled mortality risk for life annuities through **analytical annuity pricing** at a 4% interest rate
  • Performed 10,000 **Monte Carlo simulations**, with the simulated mean within 0.083% of the analytical expected present value
  • Quantified **longevity sensitivity**: a 9% mortality reduction increased benefit liability from $91,576 to $93,523 per policy

- **Machine Learning-Based Evaluation of Small and Micro Business Projects, East China Normal University**
  *Shanghai, China* Sep. 2023 – Dec. 2023
  • Structured expert reviews from 314 articles across six evaluation dimensions through **data encoding and manual annotation**
  • Developed an **MLP model in R (RSNNS)** to predict project investment recommendations, achieving 81.8% validation accuracy
  • **Cleaned data and filtered outliers**, then partitioned training and validation sets to evaluate model performance

## Professional Experience

- **Procurement & Strategic Operations (Co-op), Municipal Property Assessment Corporation (MPAC)**
  *Pickering, Canada* Apr. 2025 – Aug. 2025
  • Built an **Excel compliance dashboard** for 120+ suppliers, automating insurance expiry alerts and cutting follow-ups by 75%
  • Co-led a workflow survey across 6 business units, using **descriptive analytics** to identify bottlenecks and improve collaboration
  • Integrated EDIA and Code of Conduct disclosures into a **supplier risk dataset** to support **risk heatmapping** and tracking

- **Insurance and Enterprise Risk (Co-op), Toronto District School Board (TDSB)**
  *North York, Canada* May 2024 – Dec. 2024
  • Automated tracking for 200+ vendors with **Excel formulas**, reducing manual work across onboarding, renewals, and approvals
  • Built **Excel VBA** to classify email inquiries and summarize recurring issues, improving response efficiency and transparency
  • Maintained the Excursion website using **HTML/CSS**, resolving technical issues and redesigning layout to improve usability

- **Web Assistant (Co-op), University of Waterloo**
  *Waterloo, Canada* Sep. 2022 – Dec. 2022 & May 2023 – Aug. 2023
  • Modernized 100+ pages (Work-Learn Institute), including **link validation**, **layout redesign**, and bug fixes using **HTML**/**CSS**
  • Built responsive pages from wireframes and refined layouts for Centre for Work-Integrated Learning in a 6-person team
  • Automated web data collection with **Python** and used **Google Analytics** to assess engagement and guide website updates